1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L20 – Case Study (DFD+ERD)**

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4 **Recap**

- In the previous lecture, we learned about DFD + ER
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- Today we shall go through a case study

5 **The Case**

6 **Calendar App Case**

- A calendar app – introduced earlier
  - Recap - we are interested to build a calendar app. It is meant for the students (mainly) to help in their various academic activities

7 **SRS**

- Already discussed earlier
- Remember, SRS is the starting point for code design (our concern here)

8 **SRS – Recap (Wk 3, L8)**

9 **Note**

- Only functional requirements considered
- Inclusion of non-functional (usability) requirements as functional not considered – for simplicity

10 **DFD**

11 **DFD - Recap**

- Represents "flow of data" through a process or a system
  - Focus on data *"movement"* between external entities and processes, and between processes and data stores
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12 **Recap - Components of DFD**

- Source/Sinks (external entity)
- Processes
- Data stores
- Data flows

13 **Recap - Component Representation**

14 **Recap - DFD Levels**
- At least 3
  - Level 0 – context diagram
  - Level 1 – overview diagram
  - Level 2 – detailed diagram
- There can be further levels, if required

15 **Level 0 – Calendar App**

16 **Level 1**

17 **Level 1**

18 **Level 1**

19 **Level 1**

20 **Level 2, Process 1.0 (Display Calendar)**

21 **Level 2, Process 2.0 (Reminder Setting)**

22 **Level 2, Process 3.0 (Set Background)**

23 **Level 2, Process 4.0 (Synchronization)**

24 **Note**
- This is only ONE design, out of many possibilities
- Same SRS can be converted to different designs
  - Depending on the skill of the designer(s)
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25 **ER Diagram**

26 **Recap - Representing Data Stores**
- ERD used to "express" rich internal structure, organization and relationships in data stores

27 **Recap - Basic Components**

28 **Recap - Modeling**
- A data store (database) can be modeled as:
  - A collection of entities
  - Relationship among entities
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29 **ERD - Calendar App**

30 **Thank You**